

Coauthor Writing Guide - UPC

A paper should allow readers to quickly discover the main results, and then, if interested, to examine the supporting evidence. Structure your paper to support this behavior:

- 1) Describe the work in the context of accepted scientific knowledge.
- 2) State the idea that is being investigated, often as a theory or hypothesis.
- 3) Explain what is new about the idea, what is being evaluated, or what contribution the paper is making.
- 4) Justify the theory, by proofs or experiments.

I. A SIMPLIFIED PICTURE

- 1) Work on a (relevant) Computer Science/Software Engineering/Information System question
- 2) Write a scientific paper
- 3) Submit the paper to an appropriate journal/conference
- 4) If accepted for publication then

- Add one line to CV
- Present work at scientific meetings

- 5) Else (paper rejected or to be modified) Go to Step 2

Table I shows the activities week by week for developing:

- Conference paper
- Journal paper

II. TP1

Formats

Most of the conferences use one of these formats: [IEEE template](#), [Springer template](#), [ACM template](#), [Scitepress template](#).

Week 1

- 1) The coauthor explains the paper structure:
 - Title and information about authors
 - Abstract
 - Introduction
 - Related Works
 - Main Contribution
 - Context/Background/Overview
 - Method
 - Experiments
 - Conclusions and Perspectives
 - Bibliography

Week 2

- The students present a first version of the introduction:
 - What is the problem?
 - Why is it interesting and important?
 - Why is it hard? (e.g., why do naive approaches fail?)

- Why hasn't it been solved before? (or, what's wrong with previous proposed solutions? How does ours differ?)
- What are the key components of our approach and results? Also include any specific limitations.

Fig 1 shows the structure of the introduction in the first two pages (in red).

Week 3

- The students present a final version of the introduction, after the observations of the coauthor.

Reproducing arm movements based on Pose Estimation with robot programming by demonstration

FirstName LastName FirstName LastName FirstName LastName
Universidad Politécnica de Universidad Politécnica de Universidad Politécnica de
Cataluña (UPC) Cataluña (UPC) Cataluña (UPC)
Lluis.Pera Lluis.Pera Lluis.Pera
l.pera@upc.edu l.pera@upc.edu l.pera@upc.edu

Abstract—The teaching of robot movements has always been considered a complex subject in which there is a lot of interest, since the slightest change in the robot programming can generate a high dimension that can be even more than a month. In this work, we carried out a study of human movements to implement a new method of robot programming by demonstration using neural networks. Current methods require variables with high mathematical and logical knowledge to teach robot movements. Using a human pose estimation algorithm called OpenPose, we used a 3D filtering method to obtain the estimated pose of the person arm in a standard 3D space. Then, we use various classification tools to transfer the predicted pose into the robot. The results show that it is feasible to make robot programming more accessible using pose estimation.

Index Terms—Pose Estimation, Robot Programming by Demonstration, Open Pose, Neural Networks.

1. INTRODUCTION

Currently, many companies use robots with Machine Learning technology to perform a variety of tasks, such as welding, machine case, material control or other activity, since it would help humans to work together with robots, in order to enhance the operation of its warehouses and factories. After the emergence of the pandemic caused by COVID-19 in 2020, multiple companies are re-evaluating their business models to adapt to this situation, which has accelerated the introduction of robots in various industries, since this virus has exposed the limitations of human labor. The method currently used to establish the movements of a robot is quite complex and has a complex learning curve, since any alteration in robots would mean weeks or months of programming by people who require high knowledge, which generates a robot downtime, a high cost and low consumption time for this company [1].

There are some solutions that have a similar approach, since they seek to do robot programming by demonstration. For example, in [2], the authors are inspired by the Evolutionary Algorithms framework to propose a framework, called by them as EPG. Therefore, they analyze the "policy" to produce movement through trial and error, rather than using a pose estimation method. This allows robots to perform movements learned by human demonstration in the face of possible changes in the environment. However, this process is relatively slow, which generates an unnecessary waste of time in cases where there are no changes in the environment. In [3], the authors use the YOLO and OpenPose algorithms to get the action that is performed and records the results with a Gaussian filter, without using our estimating human pose. Instead, we estimate the human pose with OpenPose algorithm and then use a neural network to help to adapt the movement so that it can be reproduced by the robot.

The key components of our research are compared by OpenPose, the most popular free non-commercial pose estimation method used to obtain the 2D pose of the person. Then, we use the 3D filtering method in order to transfer the estimated 2D pose to a 3D pose. This method of 3D pose estimation is fast but has some precision problems with the estimated points in the same axis of the point of view. Furthermore, we use the Unity3D to generate a dataset of 3D models and the ROS integration to simulate a robotic arm that repeats the human movements.

In [4], the authors propose a new LBD (Learning from Demonstration) system that learns to perform the actions demonstrated through the recognition of patterns and objects in an experimental environment. They use the YOLO and OpenPose algorithms to obtain the action that is performed and smooth the results with a Gaussian filter, without using or estimating the human pose. Instead, we estimate the human pose and then use a neural network to convert human arm movements into robotic arm movements that represent the same action.

Our main contributions are as follows:

- We are implementing a method that allows us to transfer from the space of a video into a 3D arm estimation. Then, we transfer this information through our programming.
- We propose a new movement dataset based on the theory of the kinematics [5] using Unity3D and different person models extracted from Mousel.
- We build a test dataset which consists values of 10 test subjects performing 10 pre-set movements to be used in experiments.

This paper is structured in the following sections: First, we review the related works of robot programming by demonstration and pose estimation in Section II. Then, we talk about relevant concept and theory related to the background of our research in Section III and describe in more detail our main contribution in Section III. Furthermore, we will explain procedures performed and the experiments that were carried out in this work in Section IV. In the end, we will show the main conclusions of the project and we will indicate some recommendations for future work in Section V.

The field of Robot Programming by Demonstration has been around for a few years as people wanted to discover new ways to teach a robot movements without the need for expensive equipment like detectors. Thus, a great variety of methods related to this field were proposed over the years to improve the results more and more. In this vision, related works that use techniques similar to the ones we use are mentioned. These articles mention the disadvantages of relying on kinematic demonstration when you want to teach a robot such as the fact that robots can only learn movements for specific cases and situations, which causes them to make mistakes before various required changes in movement.

In [10], the authors propose a method that learns the movements in a kinematic way and receives visual support to know if there are deviations in the environment. This allows the robot to make movements at the face of possible changes in the environment. Instead, we use a visual aid to estimate the pose and then be able to teach the robot that movement, however, we do not adjust the possible changes in the environment, since this process would considerably increase the complexity and make the process relatively slower, which generates an unnecessary waste of time in cases where there are no changes in the environment.

Fig. 1: Introduction example

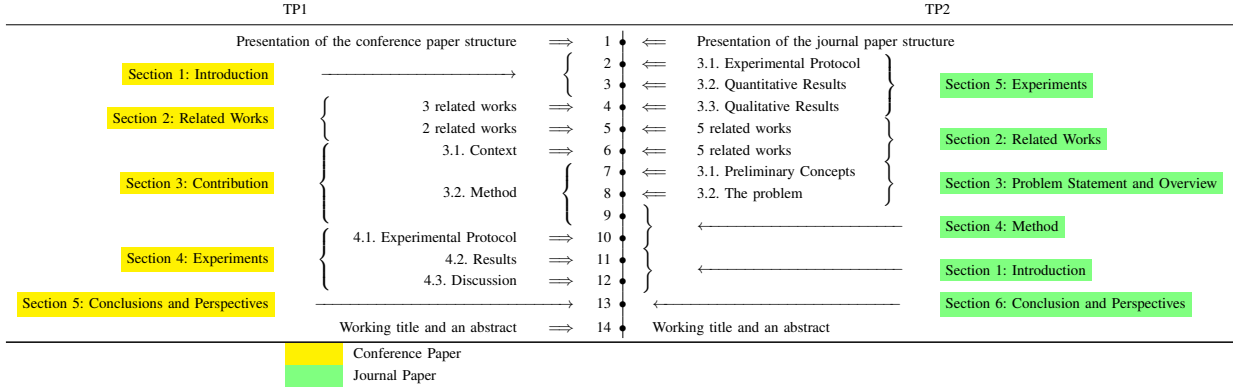


TABLE I: A Simplified Picture.

Week 4

- The students present the first version of the related works:
 - a mini-version of the state-of-the-art
 - compare your work with at least **5 papers** out of the papers from the State-of-the-Art.
 - Most results are additions to existing knowledge.
 - A literature review is used to:
 - * describe existing knowledge,
 - * compare the new results to similar previously published results, and
 - * explain how the new results extend existing knowledge.
 - Can also be used to explain how existing methods differ from one another and what their respective strengths and weaknesses are.

Fig 2 shows the structure of the related works in two pages (in blue).

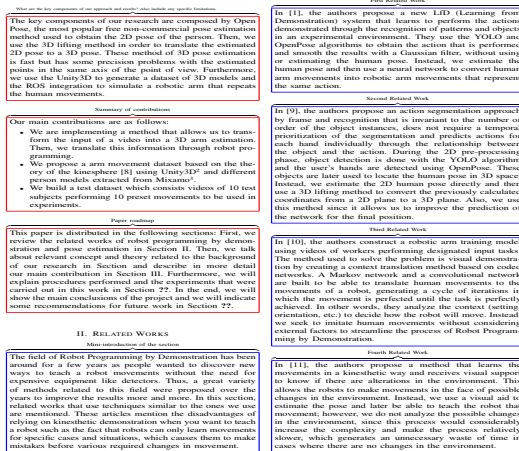


Fig. 2: Related works example

Week 5

- The students present a final version of the related works, after the observations of the coauthor.

Week 6

- The students present a final version of the context:
 - a mini-version of the theoretical framework
 - each concept/definition/theorem should have a reference
 - Provides necessary (formal) background and terminology.
 - Defines the hypothesis and major concepts.

Fig 3 shows the structure of the context in two pages (in green).

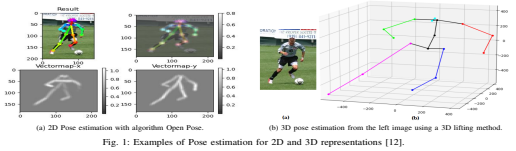


Fig. 3: Context example

- The students present a first version of the method:
 - describe in great detail your method/algorithm/model
 - put graphics/examples/tables to help the reader
 - step-by-step in the most simple and thoughtful way possible (don't make the reviewer think!!)
 - Explains the chain of reasoning that leads to the results.
 - * Provides the details of central proofs.
 - * Explains the experimental setup and summarizes the outcomes.
 - Figures, Charts, Tables, Diagrams, etc. can be extremely helpful to communicate ideas, observations, or data to others.
 - Many scientists outline their method by deciding on what figures, graphs and tables they need in order to convey their story, and then fill the text around these figures.
 - If a figure is reproduced or copied or adapted from another source, that source must be properly acknowledged in the caption, and listed among other references.

Fig 4 shows the structure of the main contribution in two pages (in red).

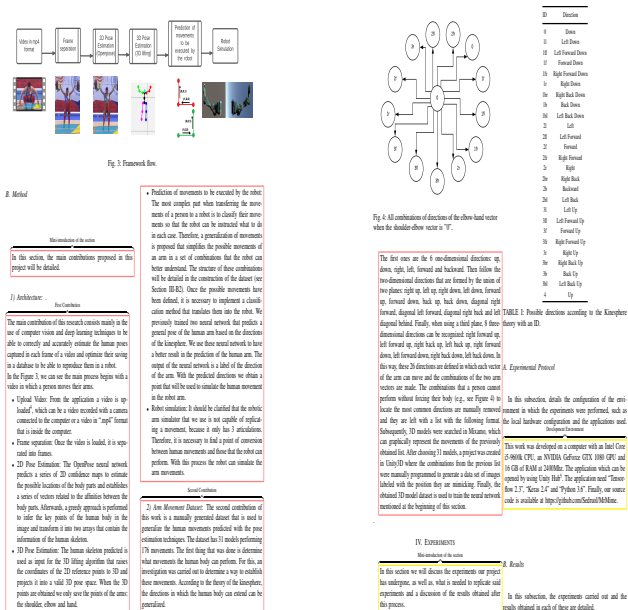


Fig. 4: Main Contribution example

- The students present a first version of the experiments:
 - States in detail and analyses the results of the research.
 - The structure should be evident in the section headings.
 - There must be a subsection called “Experimental Protocol” where it's explained:
 - * The PC configuration (RAM, CPU, ...) or the service configuration (Colab, AWS, Azure, ...)
 - * The data used (origin, construction, treatment, ...)
 - * it's better if an url with the code, or access to the service (website, demo, ...) is put here.
 - Theoretical evaluation
 - * Performance analysis
 - * Properties, complexity, etc.
 - Experimental comparative evaluation
 - * State clearly the target of experiments
 - * Experiment design: should be fair
 - * Draw conclusions from experiment results

Fig 5 shows the structure of the experiments in two pages (in yellow).

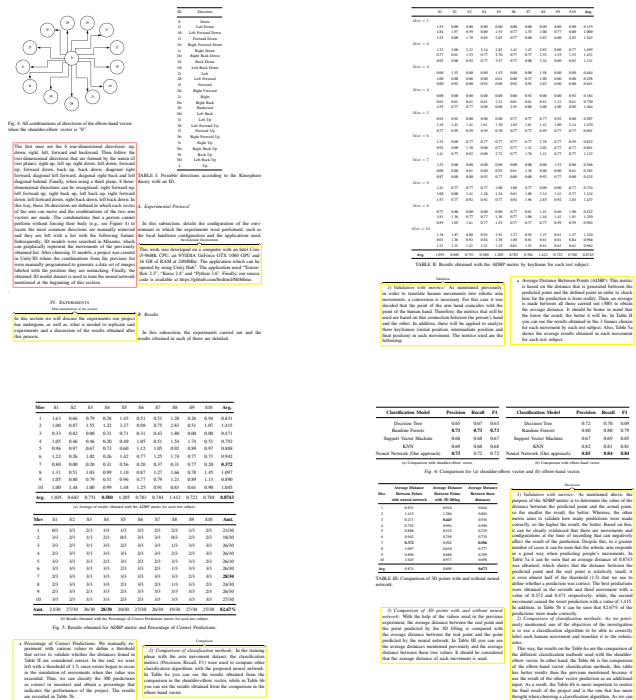


Fig. 5: Experiments example

- The students present a final version of the main contribution, after the observations of the coauthor.

- The students present a revised version of the experiments, after the observations of the coauthor.

Week 12

- The students present a final version of the experiments.

Week 13

- The students present a final version of the conclusions and perspectives:
 - Are used to draw together the topics discussed in the paper.
 - Should include a concise statement of the paper's important results and an explanation of their significance.
 - Are an appropriate place to (re)state any limitations.
 - Should look beyond the current context to:
 - * other problems that were not addressed;
 - * questions that were not answered;
 - * variations that could also be explored.
 - Perspectives should show possible future works, and show your interest in continuing with this topic.

Fig 6 shows the structure of the conclusions in two pages (in blue).

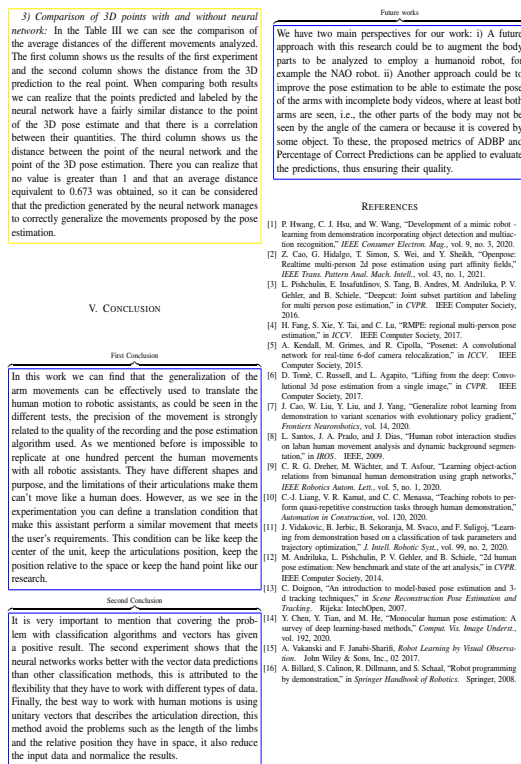


Fig. 6: Conclusions example

Week 14

- The students present a working title and a proposal of the abstract:

- Title and information about authors
- Abstract (see Fig 7):
 - * Is typically a single paragraph of 50-200 words.
 - * Allows readers to judge the relevance or the paper to them.
 - * Is a concise summary of the paper's aims, scope, and conclusions.
 - * Should be as short as possible while remaining clear and informative.
 - * The more specific, the more interesting.
 - * Self-contained and written for as broad a readership as possible.
 - * Use past tense, since it refers to work already done.
 - * **Not to put in an abstract:** minor details, paper structure, acronyms, abbreviations, mathematics, citations.

CHI 2006 Proceedings • Activity: Design Implications

Support for Activity-Based Computing Op

Jakob E. Bardram, Jonathan Bun
Centre for Pervasive Health
Aabogade 34, 8200 .
{bardram,jbp,mad

ABSTRACT

Research has shown that computers are notoriously bad at supporting the management of parallel activities and interruptions, and that mobility increases the severity and scope of these problems. This paper presents *activity-based computing* (ABC) which supplements the prevalent data- and application-oriented computing paradigm with technologies for handling multiple, parallel and mobile work activities. We present the design and implementation of ABC support embedded in the Windows XP operating system. This includes replacing the Windows Taskbar with an Activity Bar, support for handling Windows applications, a zoomable user interface, and support for moving activities across different computers. We report an evaluation of this Windows XP ABC system which is based on a multi-method approach, where perceived ease-of-use and usefulness was evaluated together with rich interview material. This evaluation showed that users found the ABC XP extension easy to use and likely to be useful in their own work.

Background

Innovation

Problem

Method

Results

Fig. 7: Abstract example

- The coauthor gives his approval of the final draft.

7. Experiments

We report an experimental study on several UCC benchmarks (see Section 7.1) and we show and discuss the practical use of our approach in a case study from cheminformatics (see Section 7.2). All experiments were conducted on a computer running Linux operating system with a core i7 processor at 2.53 GHz and a RAM of 4 GB. The implementation of Arhenius refers to [1]. The implementation of CP-Sky was carried out as described by extending the CSP-based pattern structure development [42]. For each method, reported CPU times include two steps. With Arhenius, computing the set of closed patterns was. If we use a minimal absolute frequency threshold of 1 and then applying the Sky operator. With CP-Sky, computing the set of candidates using Dynamic CSP and then filtering all patterns that are not disjunctive.

7.1. Experiments on UCC benchmarks

We chose to focus experiments on UCC benchmarks on a thoroughly investigation of the behavior of Arhenius and CP-Sky. We evaluate CPU times and several other parameters such as the number of critical patterns.

Experimental protocol. Experiments were carried out on 22 various (in terms of dimension and density) datasets (see the full column in Table 5). We considered the set of measures $M = \{freq, max, arcc, mean, growth-rate\}$. In the following, we use 6 sets of measures: the first combination of 4 measures from M ($M_1 = \{freq, arcc, mean, growth-rate\}$), $M_2 = \{freq, max, arcc, growth-rate\}$, $M_3 = \{freq, max, arcc, mean\}$, $M_4 = \{freq, max, mean, growth-rate\}$, $M_5 = \{max, arcc, mean, growth-rate\}$ and $M_6 = M$. Measures using numeric values, like $arcc$ or $mean$, were applied on attribute values that were randomly generated within the range $[0, 1]$.

<http://www.gemini.org>

<http://www.cis.uic.edu/~shankar/MSRrepository.html>

28

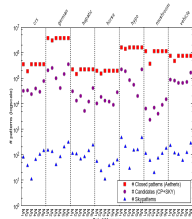


Figure 7: Computing 4 closed patterns, candidates and disjuncts in 7 UCC datasets for $M_1, \dots, M_6$.

7.2. Case study: discovering isocypines

A major issue in cheminformatics is to establish relationships between chemical values and their activity in toxicology. Chemical fragments¹ which have toxicity are called isocypines and their discovery is in the case of predictive models in toxicology [8, 52]. The aim of this study, which is part of a larger research collaboration with the CERN Lab, is to investigate the use of algorithms for discovering isocypines.

¹A fragment denotes a connected part of a chemical structure having at least one chemical bond.

32

Dataset	Dimension	Density	CP-Sky	Arhenius	CP-Sky	Arhenius	CP-Sky	Arhenius	CP-Sky	Arhenius
H400	1000	0.01	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
H401	1000	0.01	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
H402	1000	0.01	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
H403	1000	0.01	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
H404	1000	0.01	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
H405	1000	0.01	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
H406	1000	0.01	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
H407	1000	0.01	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
H408	1000	0.01	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
H409	1000	0.01	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
H410	1000	0.01	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
H411	1000	0.01	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
H412	1000	0.01	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
H413	1000	0.01	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
H414	1000	0.01	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
H415	1000	0.01	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
H416	1000	0.01	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
H417	1000	0.01	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
H418	1000	0.01	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
H419	1000	0.01	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
H420	1000	0.01	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
H421	1000	0.01	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
H422	1000	0.01	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00

Table 5: Computing the two methods on 22 UCC datasets (detailed summary).

Performance analysis. Table 5 provides the results of CPU-times for CP-Sky and Arhenius. We report for each dataset and for each set of measures M_i the number of disjuncts (indicated by the same for both methods), the number of candidates and the associated CPU-time.

¹Reported value in column 8th is associated to M_1 . For reported values in column 9th-10th reported average values over M_2, M_3, M_4, M_5, M_6 and M_7 .

29

Experimental protocol. The dataset is collected from the TCC web site¹. For each chemical, the chemical structure is with hashed standard codes (HSC) in 3 categories: H400 (very toxic, $CL50 < 1$ mg/L), H401 (toxic, $1 < CL50 < 10$ mg/L), and H402 (harmful, $10 < CL50 < 100$ mg/L). We focus on the H400 and H402 classes. The dataset T consists of 567 chemicals (transitions), 372 from the H400 class and 195 from the H402 class. The chemicals are encoded using 100 frequent closed subgraphs (patterns previously extracted) with a 75 relative frequency threshold. Therefore, the extracted disjuncts are made of these frequent closed subgraphs corresponding to chemical fragments. Discovering candidate isocypines is linked to a classification problem: we want the HSC and thus we use the growth-rate as a control measure [53]. Indeed, when a pattern has a frequency which significantly increases from the H402 class to the H400 class, then it is a potential structural alert related to an increase of the toxicity. If a chemical has, in its structure, fragments that are related to an effect, then it is more likely to be toxic. Emerging patterns embody this natural idea by using the growth-rate measure. On the other hand, real-world datasets are often noisy and patterns with low frequency may be artifacts. Thus we add the frequency measure to ensure the representativeness of the patterns (i.e., the higher the frequency the better).

The algorithm framework enables to integrate measures coming from the background domain. In toxicology, chemists know that the aromaticity measure is a chemical property that favors toxicity since their metabolites can lead to very reactive species which can interact with biomolecules in a harmful way (the higher the aromaticity, the higher the toxicity hypothesis). The chemical knowledge provides the aromaticity of the chemical fragments and we compare the aromaticity of patterns in the mean of the aromaticity of its chemical fragments.

We tested several combinations of measures: $M_1 = \{growth-rate, freq\}$, $M_2 = \{growth-rate, aromaticity\}$, $M_3 = \{freq, aromaticity\}$ and $M_4 = \{growth-rate, freq, aromaticity\}$.

¹Source: Chemical Bench, <http://www.chembench.org/>

²A chemical C is contained in T as a A/C support. A and B is a frequent subgraph of T .

33

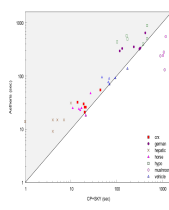


Figure 5: Computing CPU times on the 7 selected datasets for $M_1, \dots, M_6$.

Table 5 indicates that CP-Sky and Arhenius run within the same order of magnitude. On the datasets, CPU times both for CP-Sky and Arhenius are very small (less than 3 seconds), then we decided to focus on the 7 remaining datasets. For these 7 datasets, Figure 5 depicts a scatter plot of CPU-times for CP-Sky and Arhenius. Each point represents a disjunct query for a dataset. Its x-value (y-value) is the CPU-time obtained with CP-Sky (Arhenius) in y-value (x-value) scale. The plot shows a strong positive correlation between the two methods. A specific color is associated to each dataset. The line $y = x$ highlights same CPU-times. As most of the points are above this line, it means a longer time for Arhenius. With all the measures (i.e., M_1), the speedup is 1.3 (resp. 1.5) on $\{type, growth\}$ (resp. $\{freq, growth\}$). The only exception is M_6 , where the dataset has the lowest density (around 1%). We worked on the impact of the density on the CPU times. For that purpose, we generated several datasets by varying their density from 0.15 to 0.65 (as we were curious on the result on M_6), we kept the numbers of items and transactions of the dataset. Figure 6 shows the evolution of CPU times according to the density for both methods. As the items in the data are randomly generated to provide the density value, the number of closed patterns increases according to the density.

30

Measure	CP-Sky		Arhenius	
	Time	Disjuncts	Time	Disjuncts
M_1	1.00	1.00	1.00	1.00
M_2	1.00	1.00	1.00	1.00
M_3	1.00	1.00	1.00	1.00
M_4	1.00	1.00	1.00	1.00
M_5	1.00	1.00	1.00	1.00
M_6	1.00	1.00	1.00	1.00
M_7	1.00	1.00	1.00	1.00

Table 6: Skypines mining on UCC dataset.

Performance analysis. Table 6 reports, for each set of measures M_i ($i \in \{1, \dots, 6\}$) the number of disjuncts, (ii) for CP-Sky, the number of candidates and the associated CPU-time and (iii) for Arhenius, the number of closed patterns and the associated CPU-time. CP-Sky outperforms Arhenius in terms of CPU times. Moreover, the number of candidates for Arhenius is drastically smaller than the number of closed patterns computed by Arhenius. It clearly shows the interest of the filtering of the dynamically provided constraints.

Qualitative analysis. We now analyze qualitatively the skypines by comparing them with well-known environmental isocypines [55]. With the growth rate and frequency measures (i.e., M_1), only 8 disjuncts are found, and with known isocypines are emphasized. Figure 6 depicts these disjuncts derived from M_1 , one of them is on the y-axis. Two of them are components of widespread pesticides, namely the chlorinated aromatic ring ($C_6H_4Cl_2$) and organophosphorus moiety ($P(=O)(OR)_2$). The third one, the phenyl ring (C_6H_5) is related to hydrophobicity and formation of free radicals [56].

The most interesting results are provided by adding the background knowledge. Indeed, adding the aromaticity measure leads to disjuncts with novel chemical characteristics. We comment the results obtained with the growth-rate, frequency and aromaticity measures (i.e., M_2). Once again, the whole set of disjuncts remains small and can lead to a straightforward analysis. 21 disjuncts are mined (see Figure 8b). To simplify the picture, S denotes a set of disjuncts during a common chemical feature.

34

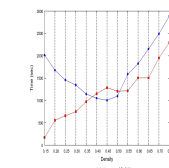


Figure 6: Measuring the impact of the density for the two methods.

Thus, the running time of Arhenius also increases according to this parameter. The behavior of CP-Sky is more complex due to the dynamic pruning. Having more candidate disjuncts is a benefit if these patterns are able to prune significantly the search space. The experimental study shows a good trade-off with a density close to 0.5. Moreover, note that in practice there are very few datasets with a density higher than 0.5.

Another interesting result provided by Table 5 is the number of closed patterns with Arhenius was the number of candidates with CP-Sky, in other words the number of representative patterns for both methods. The number of candidates with CP-Sky remains small (around 1000) compared to the number of closed patterns with Arhenius (about millions). Figure 7 summarizes these results under a graphical way. It reports for each set of measures M_i ($i \in \{1, \dots, 6\}$) and the 7 datasets investigated in Figure 1, the number of closed patterns (Arhenius), the number of candidate patterns (CP-Sky) and the number of disjuncts (both methods). It clearly illustrates the lower number of representative patterns required by CP-Sky than Arhenius.

31

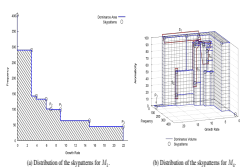


Figure 8: Analyzing the skypines.

The figure emphasizes several environmentally hazardous chemical fragments: the phenyl ring (C_6H_5), the chlorinated aromatic ring ($C_6H_4Cl_2$), the chlorinated benzene (C_6H_4Cl), and the organophosphorus moiety ($P(=O)(OR)_2$). Besides, information dealing with nitrogen aromatic compounds are also extracted (C_5H_4N). The comparison of this list of patterns with emerging fragments (EFs) extracted from previous experiments [55] highlights the generalization potency of the skypines. As an example, the organophosphorus moiety disjunct is a generalization of around 90 EFs (can be seen as a kind of maximum common structure (i.e., consensus structure) of these fragments).

Note that adding p -value as a measure in order to provide a statistical significance in the disjuncts does not significantly change the results in this experiment. Indeed, p -values estimated by the closed patterns algorithm in frequency and with the set of measures $M_2 = \{growth-rate, freq, aromaticity, growth\}$ only the aromaticity leads to another closed pattern. Out of curiosity, we ran the experiment with M_2 and we got 21 disjuncts instead of 21 with M_2 , the 21 disjuncts extracted with M_2 are still disjuncts with M_2 . There is no significant new insights from a chemical point of view.

35

Fig. 8: Experiments example

III. TP2

A. Formats

Most of the conferences use one of these formats: [IEEE template](#), [Springer template](#), [ACM template](#), or others.

Week 1

- The coauthor explain the journal paper structure:
 - Title and information about authors
 - Abstract
 - Introduction
 - Related Works
 - Problem Statement and Overview
 - Method
 - Experiments

Week 2

- The students present a first version of the Experiments section: a in depth version of the Experiments of the conference paper (see TP1 Weeks 10-12)
- Fig 8 show the structure of the Results (in blue).

Week 3

- The students present a revised version of the Results, after the observations of the coauthor.

Week 4

- The students present a final version of the Results.

2. Related work

2.1. Pattern mining

Frequent itemset mining was first described in [10]. The problem can be defined as follows: a transaction is a subset of a given set of items $\mathcal{I}$, and a transaction database, denoted $\mathcal{T}$, is a set of such transactions. A subset X of $\mathcal{I}$ is a frequent itemset in $\mathcal{T}$ if the number of transactions containing X exceeds a given threshold, denoted by σ . One of the earliest findings in the data mining community is that a mining process usually produces large collections of patterns. Many researchers have proposed methods to reduce the size of the output. The constraint-based pattern mining framework [30], the condensed representations [29] and the compression of the dataset by exploiting Minimum Description Length Principle [31], to name a few. A general observation is that patterns represent “fragmented knowledge”, and often there is no clear view of how these knowledge fragments interact and combine to produce a global model. Recent approaches have therefore used schemes such as pattern teams [32], constraint-based pattern set mining [33] and pattern selections [34] that aim to minimize the redundancy and the number of patterns. A common theme in these studies is to select patterns from the initial large set of patterns on the basis of their usefulness in a given context. This approach comes within the general trend to produce pattern sets i.e. sets of patterns satisfying properties on the whole set of patterns [33]. Other approaches take advantage of closed patterns to maximize a specific measure such as the growth rate for emerging patterns [35] and the area for tiling [36, 14]. Often, these methods focus on optimizing a global measure on the discovered pattern set and neglect the relationships between patterns. Moreover, these approaches suffer from a lack of flexibility to express the queries requested by the analyst. For each method, the user has to understand its semantics and express queries satisfying its algorithmic properties and constraints.

Another class of techniques considers statistical significance of patterns. The objective is to extract patterns for which a given characteristic (usually the frequency) deviates so much from its expected value under a null model that it is unlikely to have been generated by it. The frequency of a pattern is then considered as a random variable, whose distribution under the null hypothesis has to be calculated or approximated, and the significance of the pattern is assessed through a statistical test that compares the expected frequency under the null model to the observed frequency.

The null model will ideally involve a *trade-off* between adjustment to the data and simplicity. The model should capture some characteristics of the data, otherwise all patterns would be significant, but it should not overfit the data. A number of works have explored various notions of statistical significance for itemsets and have proposed novel and efficient methods for their extraction. In [37], an approach is developed to mine statistically significant association rules, under an assumption of independence between the condition and the consequence of the rule. The expected frequency for the condition and consequence is derived analytically, based on the marginal frequencies of the dataset. In [20], the authors use two null models to calculate the expected frequencies. The first is an independence model, where items are assumed to be mutually independent and their probabilities of occurrence are their observed marginal relative frequencies. The second model is obtained by assuming a distinct distribution for each transaction, where each of the items has the same transaction-dependent probability of occurrence. This second model considers large transactions to be less interesting, because two such transactions are likely to share items by chance. As the expected frequency is anti-monotone (the expected frequency of an itemset is greater than the expected frequencies of its subsets), the authors only consider closed itemsets (itemsets having no super-set with the same observed frequency). They then introduce the MIN algorithm to mine for non-redundant unexpected itemsets, by penalizing itemsets overlapping with itemsets with the highest p -values. An alternative to the independence model is the maximum entropy model [38]. A related framework is developed in [39], where unexpected itemsets with respect to a maximum entropy model are used to update the model iteratively, resulting in a large coverage of the database and minimal redundancy between the itemsets.

Pattern mining and Constraint programming. Pattern mining benefits from the recent progress on cross-fertilization between data mining and Constraint Programming (CP) [40, 41, 42, 25]. CP is a general declarative methodology for solving constraint satisfaction problems. Within this framework, the user specifies in a declarative way by using constraints what the problem is rather than a method dedicated to the problem. Then a general solver provides the complete set of solutions satisfying all the constraints. Results show that these methods are more expressive and work well.

2.2. Skyline

The notion of *dominance* is at the core of the skyline processing. This notion can also be used to handle many pattern mining problems that involved pairwise comparisons between patterns [3]. The skyline points can be viewed as *compromise points* with respect to a given set of criteria. Skyline computation is strongly related to mathematical and microeconomics problems such as maximum vectors [43], Pareto set [44], and multi-objective optimization [45]. Since its rediscovery within the database community by Börzsönyi *et al.* [28], many methods have been developed for answering skyline queries that can handle various constraints in different computational environments [46, 47]. Skyline queries focus on the extraction of tuples from a given dataset and assume that all the elements are in the dataset, while the skyline mining task consists of extracting patterns which are elements of the frontier defined by the given measures. The skyline problem is clearly harder because the search space for skylines is much larger than the search space for skylines (cf. Section 3.2). Another aspect of preference-based processing is the *top-k* procedure [26, 27]. A ranking function f_i is applied to patterns, and the k best patterns with the highest score with respect to f_i are returned. As previously mentioned, this approach suffers from some limitations. The choice of k is not trivial (i.e., the horizon problem). A low value may miss useful patterns and too high a value introduces redundancy within the produced patterns (i.e., highly similar patterns). This limitation is the main motivation for the notion of the “most informative patterns” (MIP) that were proposed in [48]. MIPs can be seen as patterns that *locally dominate* other patterns according to a scoring function. This approach shares a similar spirit to ours as it also limits the number of enumerated patterns to a more manageable level. However, in contrast to our approach, work on MIPs includes a notion of dominance that is *local and specific* only to subsets of patterns.

Fig. 9: Related Works example

Week 5

- The students present a final version of the Related Works:
 - a mini-version of the state-of-the-art
 - compare your work with **10 papers** out of the papers from the State-of-the-Art.
 - Most results are additions to existing knowledge.

Fig 9 shows the structure of the related works in two

pages (in blue).

Week 6

- The students present a final version of the Related Works.

6. Missing algorithms using Dynamic CSP

This section describes our dynamic method called CHS for the algorithms missing problem. We show how this problem can be modeled and solved by using Dynamic CSP [5]. A major advantage of this method is to improve the missing step during the process thanks to constraints dynamically posted and removing from the current set of the candidate solutions. These constraints avoid producing solutions dominated by the current solution and thus reduce the search space. More precisely, each time a solution is found, a new constraint is dynamically posted. The process stops when we cannot enlarge the "dominated area" (cf. Figure 3). The set of obtained representative pattern is a subset of the set of representative pattern extracted by Ackerly (cf. Section 3.3). Finally, the completeness of our method is insured by the completeness of the CSP solver. The implementation has been carried out in *Prolog*³ extending the CSP based pattern extractor developed by [42].

6.1. CSP and Dynamic CSP

This section recalls background on Constraint Satisfaction Problem and Dynamic CSP.

CSP. A Constraint Satisfaction Problem (CSP) (P, D, C) is defined by:

- a finite set of variables $X = \{x_1, x_2, \dots, x_n\}$,
- a domain D_i which maps every variable $x_i \in X$ to a finite set of values $D_i(x_i)$,
- a finite set of constraints C .

Algorithm 1 [40] shows how a CSP can be solved using a depth-first search D and C' denote respectively the current domain and the current set of constraints. In each node of the search tree, the algorithm branches by assigning values to a variable that is unified (line 7). It backtracks when a violation of constraints is found, i.e. at least one domain is empty (line 2). The search is further optimized by carefully choosing the variable that is first next (line 5); for instance, heuristics from *log* select the variable x having the smallest ratio between the size of its current domain and the number of constraints it occurs.

³<http://www.prolog.org/>

21

The main concept used to speed up the search is Filtering (constraint propagation) (line 1). Filtering removes the domains of variables such that the domain remains locally consistent. A solution is obtained (line 9) when each domain $D_i(x_i)$ is reduced to a singleton and all constraints are satisfied.

```

Algorithm 1: Constraint-Search( $D, C$ )
1  $D \leftarrow propagate(D, C)$ ;
2 if there exists  $x_i \in X$  s.t.  $D_i(x_i)$  is empty then
3   return failure;
4 if there exists  $x_i \in X$  s.t.  $|D_i(x_i)| > 1$  then
5   select  $x_i \in X$  s.t.  $|D_i(x_i)| > 1$ ;
6   forall  $v \in D_i(x_i)$  do
7      $C \leftarrow Constraint-Search(D \cup \{x_i = v\}, C)$ ;
8   else
9     output solution  $D$ ;

```

Dynamic CSP. A Dynamic CSP [5] is a sequence $P_1, P_2, \dots, P_n$ of CSP each one resulting from some changes in the definition of the previous one. These changes may affect every component in the problem definition: variables (addition or removal), domains (value additions or removals), constraints (addition or removals). Solving such dynamic CSP involves solving a single CSP with additional constraints posted during search. Each time a new solution is found, new constraints $\delta(C)$ are imposed. Such constraints will survive backtracking and state that next solutions should verify both the current set of constraints C and $\delta(C)$. See line 9 of Algorithm 1 hereafter.

```

1 Output solution  $D$ ;
2  $C \leftarrow C \cup \delta(C)$ ;

```

Note that δ is a variable which is global to all calls to the procedure *Constraint-Search*(D, C).

22

6.2. Missing algorithms using Dynamic CSP

This section provides our CSP method (CHS) for missing algorithms. As previously, the representative patterns are searched according to M' by using the dynamizability principle. The key idea is the introduction of constraints on the dominance relation that are dynamically posted during the missing process. These constraints avoid producing solutions dominated by the solutions already extracted and thus reduce the search space.

Variable x will denote the (unknown) digiters we are looking for. In our method, changes on the dynamic CSP are only performed by adding new constraints. So, we consider the sequence $P_1, P_2, \dots, P_n$ of CSP where each $P_i = (X_i, D_i, C_i)$ and:

- $q_i(x) = dom_i(x)$ denotes the representative pattern x ;
- $q_{i+1}(x) = q_i(x) \wedge \delta_i(x)$ where δ_i is the first solution to query $q_i(x)$.

First, the constraint $dom_i(x)$ states that x must be either a closed pattern w.r.t. M' (i.e. $dom_i(x) = dom_{M'}(x)$) or a free pattern w.r.t. M' (i.e. $dom_i(x) = free_{M'}(x)$). Then, the constraint $\delta_i(x) = \neg q_i(x) \vee x_p > x_q$ states that the next solution which is extracted will not be dominated by q_i . Using a short notation, *proof*, we can easily argue that query $q_{i+1}(x)$ looks for a pattern x that will not be dominated by any of the patterns $q_1, q_2, \dots, q_i$.

Each time the first solution x_i to query $q_i(x)$ is found, we dynamically post a new constraint $\delta_i(x)$ leading to reduce the search space. This process stops when we cannot enlarge the dominated area (i.e. there exists x s.t. query $q_{i+1}(x)$ has no solution). For algorithms, $\delta_i(x)$ states that $\neg q_i(x) \vee x_p > x_q$.

$$\delta_i(x) \equiv \left(\bigvee_{m \in M'} m(x) < m(x) \right) \vee \left(\bigwedge_{m \in M'} m(x) = m(x) \right)$$

But, for the extracted patterns $x_1, x_2, \dots, x_n$, we are not necessarily all algorithms. Some of them can only be "intermediate" patterns simply used to enlarge the dominated area. A post processing step must be performed to filter all candidate patterns x_i that are not digiters, i.e. for which there exists s ($2 \leq s \leq n$) s.t. x_s dominates x_i . So missing digiters is achieved in a two-steps approach:

1. Compute the set $S = \{x_1, x_2, \dots, x_n\}$ of candidates using Dynamic CSP.
2. Filter all patterns $x_i \in S$ that are not digiters.

23

While the number of candidates is small, it could be very large. A heuristic reasonably used in practice for the experiments we conducted (see the experiment section).

6.3. Pattern encoding

We now introduce the model of a pattern that can be provided to the constraint programming system. Let $A = \{a, b, c, d, e, f, g, h, i, j, k, l, m, n, o, p, q, r, s, t, u, v, w, x, y, z\}$ be the 26 letters where, for each transaction t and each item i , $t_i = 1$ if t contains i . Pattern variables are variables represented by their characteristic function with boolean variables. [48, 44] model an unknown pattern x and its associated dataset T by introducing two sets of boolean variables:

- $\{X_i | i \in A\}$ where $\{X_i = 1 | \exists t \in T, t_i = 1\}$;
- $\{T_i | i \in T\}$ where $\{T_i = 1 | \exists x \in X, x_i = 1\}$.

Each set of boolean variables aims at representing the characteristic function of the unknown patterns.

The relationship between x and T is modeled by posting related constraints stating that, for each transaction t , $\{T_i = 1 | \exists x \in X, x_i = 1\}$ is covered by x :

$$\forall t \in T, (t_i = 1) \Rightarrow \bigvee_{x \in X} \bigwedge_{i \in A} (x_i = 1 \wedge t_i = 0) = 0 \quad (1)$$

A related constraint associates n variables to a constraint reflecting whether the constraint is satisfied (value 1) or not (value 0). Such constraints are useful for expressing propositional formula over constraints and for expressing that a certain number of constraints hold. Related constraints do not enjoy the same level of propagation as simple constraints. And for the other datasets $T = \{t \in T | t_i = 0\}$, then the sum must be equal to 0 (i.e., must be different from 0). The propagation is also performed, in a way, from the constraints toward the equality constraint. Using these related constraints, some measures are easy to encode: $freq(x) = \sum_{t \in T} t_i$ and $min(x) = \sum_{t \in T} t_i$. The minimal frequency constraint $freq(x) \geq f$ where f is a threshold is encoded by the constraint $\sum_{t \in T} t_i \geq f$.

24

6.4. Closeness constraint

This section describes how to encode $dom_{M'}(x)$. As an illustration, we provide the examples of $M' = \{m, n\}$ and $M' = \{m, n, o\}$. Recalling that thanks to the dynamizability, these measures allow to handle measures such as *mean*, *proportion*, *growth rate*, etc. In practice, these two examples are enough for missing experiments given in Section 7.

Let $M' = \{m, n\}$ and $min(x)$ is a function that associates an attribute value to each item. If item i belongs to x , then its value must be greater than or equal to the min. Conversely, if its value is greater than or equal to the min, then it must belong to x (if not, x would not be minimal for inclusion). Item i belongs to x is encoded as $\{X_i = 1\}$. So, x is a closed pattern for the measure *min* if:

$$\forall i \in T, (X_i = 1) \Rightarrow min(x) \geq min(m[i]) \wedge i \in x \quad (2)$$

Let $M' = \{m, n\}$, the closeness constraint ensures that a pattern has no support with the same frequency. If item i belongs to x , it is obvious that $freq(x \cup \{i\}) = freq(x)$. Conversely, if $freq(x \cup \{i\}) = freq(x)$, then i must belong to x (if not, x would not be minimal for inclusion). $freq(x)$ is encoded as $\sum_{t \in T} t_i$ and $freq(x \cup \{i\})$ is encoded as $\sum_{t \in T} t_i + x_i$. Finally, the constraint $dom_{M'}(x)$ is encoded using Equation (2) and Equation (3).

$$\forall i \in T, (X_i = 1) \Rightarrow \sum_{t \in T} t_i \wedge (1 - X_i) = 0 \quad (3)$$

6.5. Freeness constraint

Similarly to the closeness constraint, we now give two examples of encoding of the freeness constraint. Following our principle of dealing with measures used in the experiments, we give the examples of $M' = \{m, n\}$ and $M' = \{m, n, o\}$ because applying the minimal constraint on *mean* gives *max* (cf. Table 4).

Let $M' = \{m, n\}$ and $min(x)$ is a function that associates an attribute value to each item. If item i does not belong to x , then its value must be greater than or equal to the min. Conversely, if its value is greater than or equal to the min, then it must not belong to x (if i belongs to x , it would not be minimal for inclusion). Item i does not belong to x is encoded as $\{X_i = 0\}$. So, x is a free pattern for the measure *min* if:

$$\forall i \in T, (X_i = 0) \Rightarrow min(x) \leq min(m[i]) \wedge i \notin x \quad (4)$$

25

Let $M' = \{m, n\}$, the freeness constraint ensures that a pattern has no support with the same frequency. If item i does not belong to x , it is obvious that $freq(x \cup \{i\}) = freq(x)$. Conversely, if $freq(x \cup \{i\}) = freq(x)$, then i must not belong to x (if i belongs to x , x would not be minimal for inclusion).

In order to encode $freq(x \cup \{i\})$, we introduced boolean variable T_i such that the relationship between $x \cup \{i\}$ and T is modeled by posting related constraints stating that, for each transaction t , $\{T_i = 1 | \exists x \in X, x_i = 1\}$ is covered by $x \cup \{i\}$.

$$\forall t \in T, (T_i = 1) \Rightarrow \sum_{x \in X} \bigwedge_{i \in A} (x_i = 1 \wedge T_i = 0) = 0 \quad (5)$$

$freq(x)$ is encoded as $\sum_{t \in T} t_i$ and $freq(x \cup \{i\})$ is encoded as $\sum_{t \in T} t_i + T_i$. So $freq_{M'}(x)$ is modeled using Equation (1) and Equation (6).

$$\forall i \in T, (X_i = 0) \Rightarrow \sum_{t \in T} t_i - T_i = 0 \quad (6)$$

6.6. Solving the missing example using Dynamic CSP

We now illustrate CHS on the missing example in Table 1 with $M' = \{m, n, o, p\}$. We use the maximal constraint T and thus the closeness constraint. As $T(m, n) = freq$, we get $M' = \{freq\}$. Figure 4 depicts the various steps of the resolution.

26

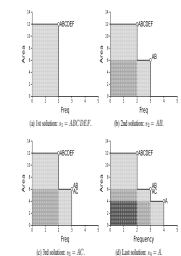


Figure 4: Solving the missing example using Dynamic CSP.

Let P be the associated dynamic CSP (see Section 6.2). $P = (\{X_i, C_i, D_i\})$ where query $q_i(x) = dom_{M'}(x)$. In the first solution is pattern $x_1 = ABCDEF$ (with $freq(x_1) = 2$ and $min(x_1) = 12$), cf. Figure 4a. So, we consider query $q_2(x) = dom_{M'}(x) \wedge \neg q_1(x) \vee x_p > x_q$ stating that we are looking for a closed pattern x neither dominated by x_1 nor x . Its first solution is pattern $x_2 = AB$ (with $freq(x_2) = 1$ and $min(x_2) = 6$), cf. Figure 4b. The next query is $q_3(x) = dom_{M'}(x) \wedge \neg q_1(x) \vee \neg q_2(x) \vee x_p > x_q$ stating that we are looking for a closed pattern x neither dominated by x_1 nor x_2 . Its first solution is pattern $x_3 = AC$ (with $freq(x_3) = 1$ and $min(x_3) = 6$), cf. Figure 4c. The next query is $q_4(x) = dom_{M'}(x) \wedge \neg q_1(x) \vee \neg q_2(x) \vee \neg q_3(x) \vee x_p > x_q$ (cf. Figure 4d) and then query $q_5(x) = dom_{M'}(x) \wedge \neg q_1(x) \vee \neg q_2(x) \vee \neg q_3(x) \vee \neg q_4(x) \vee x_p > x_q$ has no solution since the dominated area cannot be enlarged and the process ends for $i = 5$.

In this example, note that all extracted patterns are digiters (i.e., there is no intermediate pattern). The CSP system did not generate solution that does not satisfy the dominance relation. Experiments in the next section provide examples with intermediate patterns.

27

Fig. 11: Method example

Week 9

- The students present a final version of Method section: a in depth version of the main contribution of the conference paper (see TP1 Weeks 7 to 9). Fig 11 show the Method section (in green).

Week 10

- The students present a final version of the Method section, after the observations of the coauthor.

1. Introduction

The process of extracting useful patterns from data, called *pattern mining*, is an important subfield of data mining, and has been used in a wide range of applications and domains such as bioinformatics [5], cheminformatics [6], social network analysis [7], web mining [8] and network intrusions detection [9]. Since the seminal papers of Agrawal *et al.* [10], Mamola *et al.* [11], a considerable number of patterns, such as itemsets, strings, sequences, trees and graphs, have been studied and used in real-world applications. Nowadays, many pattern extraction problems like subgroup discovery [12], discriminant pattern mining [13], and tiling [14] are understood from both theoretical and computational perspectives.

Most existing pattern mining approaches enumerate patterns with respect to a given set of constraints that range from simple to complex. For instance, given a transaction database, a well-known pattern mining task is to enumerate all itemsets (i.e., sets of items) that appear in at least s transactions. Another mining approach is to extract from a given graph database all subgraphs that have a diameter larger than l , connectivity higher than c , and where each vertex has a degree bounded by d . The output for these mining approaches, however, can be extremely large even for moderately sized datasets. For instance, in the worst case, the number of frequent itemsets is exponential in the size of the items in the dataset. So far, the community has made great efforts on sophisticated algorithms which push the constraints deep into the mining process [15]. But also in condensation (i.e., reduction) techniques to limit the number of output patterns depending on the application contexts. Examples of these condensation techniques include: closed itemsets [16], non-derivable itemsets [17] and free itemsets [18]. Several other studies in the pattern mining community focused on approaches that consider a statistical null hypothesis and which assess patterns according to how much their observed frequency (or more rarely the length [19]) deviates from their expected frequency under the null model [20, 21, 22, 23]. One of the key points of such approaches is to choose an appropriate null hypothesis. It will involve a trade-off between adjustment to the data and simplicity.

The pattern mining community, however, has paid less attention to combining mining constraints. In practice, many constraints entail choosing threshold values such as the well-used minimal frequency. This notion of “thresholding” has serious drawbacks. Unless specific domain knowledge is available, the choice is often arbitrary and may lead to a very large number of extracted patterns which can reduce the success of any subsequent data analysis. This drawback is even deeper when several thresholds have to be combined. A second drawback is the *stringent enumeration aspect*: a pattern is either above or below a threshold. But what about patterns that respect only some thresholds? Should they be discarded? It is often very difficult to apply *subtle selection mechanisms*. There are very few works such as [24, 25] which propose to introduce a softness criterion into the mining process. Other studies blend user preferences in the mining task in order to limit the number of extracted patterns such as the top- k patterns [26, 27]. By associating each pattern with a *rank score*, this approach returns an ordered list of the k patterns with the highest score to the user. However, combining several measures in a single scoring function is difficult and the performance of top- k approaches are often sensitive to the size of the datasets and to the threshold value, k .

Here, we present a new framework that aims to make the results of pattern mining *useful from a user-preference point of view*. To this end, we integrate into the pattern discovery process the idea of skyline queries [28] in order to mine *skyline patterns* in a threshold-free manner. Such queries have attracted considerable attention due to their importance in multi-criteria decision making and economics where they are usually called “*Pareto efficiency or optimality queries*”. Briefly, in a multidimensional space where a preference is defined for each dimension, a point a dominates another point b if a is better (i.e., more preferred) than b in at least one dimension, and a is not worse than b on every other dimension. For example, a user selecting a set of patterns may prefer a pattern with a low frequency, a large length and a high confidence. In this case, we say that pattern a dominates another pattern b if a .frequency $\leq$ b .frequency, a .length $\geq$ b .length, a .confidence $\geq$ b .confidence, where at least one strict inequality holds. Given a set of patterns, the skyline set contains the patterns that are not dominated by any other patterns.

Skyline pattern mining is interesting for several reasons. Firstly, skyline processing does not require any threshold selection or ranking function. Secondly, the formal property of dominance satisfied by the skyline pattern defines a global interest with semantics easily understood by the user. These semantics are discussed at length in the economics literature, where the Pareto efficiency is applied to the selection of alternatives in resource distribution. However, while this notion of skyline has been extensively developed in engineering and database applications, it has remained unused for data mining purposes except for just a few works in the last three years [1, 2, 3, 4].

Contributions and roadmap

Here, we present significant extensions of our recent papers [1, 4] on the discovery of skyline patterns, or “*skypatterns*”. First, we detail the *static method* based on the theoretical relationships with pattern condensed representations (representations which return a subset of the patterns having the same expressiveness as the whole set of patterns [29]). Second, we describe the *dynamic method* which produces a continuous refinement of the skyline constraints based on the extracted patterns. This is achieved through a Dynamic CSP (Constraint Satisfaction Problems) framework. Third, the key notion of “*skyliness*” which constitutes the cornerstone of our two methods is explained in more detail. Furthermore, we discuss the *generic* aspect of the skypattern framework and we show that we can combine different measures to provide statistical assessments of pattern mining results. Finally, we present an extensive empirical study which includes a wide range of datasets and comparisons of our techniques. This enables us to draw some lessons on the strengths and weaknesses of each method.

The rest of this paper is organized as follows. Section 2 surveys the works related to skyline pattern analysis. Section 3 introduces some basic definitions, the formal problem statement and an overview of our work. The key notion of skyliness is then studied in Section 4. Section 5 discusses the computation of pattern condensed representation for skypattern queries. Section 6 discusses skyliness but within a dynamic CSP framework. We report an empirical study on several datasets and a case study from the cheminformatics domain in Section 7. Finally, Section 8 discusses the learnt lessons.

Fig. 12: Introduction example

Week 11

- The students present: A first version of the introduction, that is not the same as the one for the conference paper. It should clearly state that it is an extension of a previous paper. Fig 12 shows the structure of the introduction in the first three pages (in red).

Week 12

- The students present a final version of the introduction, after the observations of the coauthor.

8. Conclusion and Perspectives

In this paper, we investigate in detail the skyline pattern mining problem by studying the theoretical relationships between pattern condensed representations and skylines. Based on the concept of skylineability, we have devised the static method Aetheris and the dynamic method CP+Sky. Aetheris exploits the pattern condensed representations to provide a proper superset of skylines to apply the Sky operator. CP+Sky iteratively refines the skyline constraints using the extracted patterns. This gives a better pruning of the search space. We produce the complete set of skylines in a generic way (i.e., with a large set of measures that includes statistical assessments such as the p-value). The practical goal is to make the result of pattern mining useful from a *user-preference point of view*. One strength of the approach lies in the fact that no threshold has to be set, the end-user only needs to specify as input the dataset and the set of measures she is interested in.

An extensive empirical study as well as a case study from chemoinformatics show the efficiency of our two algorithms according to both quantitative and qualitative aspects. Despite the gain in generality brought by the CSP framework and the fact that Aetheris takes benefit of the pruning strategies based on the anti-monotonicity to extract patterns, CP+Sky competes with Aetheris and even outperforms it in some cases. Moreover, with CP+Sky, the search order of the patterns may significantly impact the efficiency of the CSP machinery. Investigating the most beneficial order in which patterns are produced is a promising research track to maximize the effectiveness of the strategy of the constraints dynamically posted.

Skyline mining can generally be applied to a wide range of problems by adapting either the language or the dominance relation. For instance, the language of sequences or graphs can also produce skylines. The dominance relation can be changed or extended to take into account different criteria for user-preferences. We sketch now these two issues.

Language. Our formalization, and especially the skylineability notion, is general enough to be applied to a large set of languages (sequences, trees and graph for instance). However, a change in the language can impact the efficiency of the extraction methods. Regarding Aetheris, the efficiency of the approach is based on Theorem 1 involving the distinct and indistinct operators. As aforementioned, with itemset patterns and the frequency measure, the distinct operator corresponds to the well-known closed or free frequent pattern condensed representations. Consequently, the efficient extraction of skylines for more complex languages (i.e., skyline sequential patterns or skyline graph patterns) is strongly correlated to the advances and progress on complex pattern condensed representations.

Evaluating efficiently the distinct operator on more complex patterns such as sequences, trees and graphs implies additional challenges. To cite one, in the case of sequences, convenient properties such as the *free patterns a priori property* [56], which implies effective search space pruning, cannot be used. However, there are already many methods of extraction dedicated to closed sequential patterns, closed graphs and so on. Extending CP+Sky to other languages is a challenging task due to the difficulty of modeling complex patterns. To the best of our knowledge, only certain types of pattern in sequence mining have been successfully studied [57, 58].

Dominance relation. Dominance relation is based on two levels: the measures that form basic preference criteria (e.g., frequency, area) and the combination of these basic preferences (here, the Pareto composition). Primitive-based measures are flexible enough to allow the user to express a wide variety of criteria. Indeed, classical interestingness measures for pattern mining (such as frequency, growth rate), utility functions and the most recent measures of significance (like p-value) fall within this framework. Of course, Aetheris depends again on condensed representations that are well-adapted for the desired measures. Conversely, all interestingness measures are easily expressible with CP+Sky.

Thanks to its declarative nature, CSP offers through CP+Sky a very flexible way to change the dominance relationship. For instance the strict dominance (i.e., a pattern is dominated by another pattern when the latter has a better value for all measures in M), is easily configurable with CSP and this relaxation of the dominance relation leads to the mining of soft-skylines [4].

References

- [1] A. Soulet, C. Raisi, M. Plantevit, B. Crémilleux, Mining dominant patterns in the sky, in: ICDM, 2011, pp. 655–664.
- [2] M. van Leeuwen, A. Ukkonen, Discovering skylines of subgroup sets, in: ECML/PKDD (3), 2013, pp. 272–287.
- [3] B. Nègrevergne, A. Dries, T. Gams, S. Nijssen, Dominance programming for itemset mining, in: ICDM, 2013, pp. 557–566.
- [4] W. Ugarte, P. Boizumant, S. Louhi, B. Crémilleux, A. Lepailleur, Mining

37

(a) Conclusions example.

Box 7. Nature's abstract-writing template



How to construct a Nature summary paragraph

Annotated example taken from *Nature* 435, 114–118 (5 May 2005).

One or two sentences providing a basic introduction to the field, comprehensible to a scientist in any discipline.

Two to three sentences of more detailed background, comprehensible to scientists related disciplines.

One sentence clearly stating the general problem being addressed by this particular study.

One sentence summarising the main result (with the words 'here we show' or their equivalent).

Two or three sentences explaining what the main result reveals in direct comparison to what was thought to be the case previously, or how the main result adds to previous knowledge.

One or two sentences to put the results into a more general context.

Two or three sentences to provide a broader perspective, readily comprehensible to a scientist in any discipline, may be included in the first paragraph if the editor considers that the accessibility of the paper is significantly enhanced by their inclusion. Under these circumstances, the length of the paragraph can be up to 300 words. (The above example is 190 words without the final section, and 250 words with it).

During cell division, mitotic spindles are assembled by microtubule-based motor proteins^{1–3}. The bipolar organization of spindles is essential for proper segregation of chromosomes, and requires plus-end-directed homodimeric motor proteins of the widely conserved kinesin-5 (Kif2c) family⁴. Hypotheses for bipolar spindle formation include the 'push-pull mitotic spindle' model, in which kinesin-5 and opposing motor proteins act between overlapping microtubules^{5–7}. However, the precise roles of kinesin-5 during this process are unknown. Here we show that the generate kinesin-5 Eg5 drives the sliding of microtubules depending on their relative orientation. We found in controlled *in vitro* assays that Eg5 has the remarkable capability of simultaneously moving at ~20 nm s⁻¹ towards the plus-ends of each of the two microtubules it crosslinks. For anti-parallel microtubules, this results in relative sliding at ~40 nm s⁻¹, comparable to spindle pole separation rates *in vivo*⁸. Furthermore, we found that Eg5 can tether microtubule plus-ends, suggesting an additional microtubule-binding mode for Eg5. Our results demonstrate how members of the kinesin-5 family are likely to function in mitosis, pushing apart interpolar microtubules as well as recruiting microtubules into bundles that are subsequently polarized by relative sliding. We anticipate our assay to be a starting point for more sophisticated *in vitro* models of mitotic spindles. For example, the individual and combined action of multiple mitotic motors could be tested, including minus-end-directed motors opposing Eg5 motility. Furthermore, Eg5 inhibition is a major target of anti-cancer drug development, and a well-defined and quantitative assay for motor function will be relevant for such developments.

Nature's abstract-writing template, <https://www.nature.com/nature/for-authors/formatting-guide>. © 2021 Springer Nature Limited. All rights reserved. Permission for further reuse must be obtained from the relevant holder of the exclusive rights.

(b) Abstract example.

Fig. 13: Conclusions and Abstract examples.

Week 13

- The students present a final version of the conclusions and perspectives: Fig 13a shows the structure of the conclusions in two pages (in blue).

Week 14

- The students present:
 - Title and information about authors
 - Abstract (see Fig 13b): it should clearly state that is an extended version of a previous paper.
 - The coauthor gives his approval of the final draft.
- Finally, two appendix to check the final draft.

APPENDIX

You should look for these typos in the drafts presented by students:

- Consistency:
 - Are the titles and headings consistent with the content?
 - Have all terms been defined?
 - Is the style of definition consistent? For example, were all new terms introduced in italics, or only some?
 - Has terminology been used consistently?
 - Are abbreviations and acronyms stated in full when first used? Are any abbreviations or acronyms introduced more than once?
 - Are any abbreviations used less than, say, four times? If not, can they be removed?
- Syntax:
 - Has a term been capitalized in one place and not in another?
 - Is the style and wording of headings and captions consistent?
 - Are names always used in the same way?
 - Is spelling consistent? What about “-ise” versus “-ize”, “dispatch” versus “despatch”, or “disc” versus “disk”?
 - Is tense used correctly?
 - Is the use of indentation consistent?
 - Do the parentheses match?
- References:
 - Are references discussed in a consistent way?
 - Have bold and italic been used logically?
 - In the references, has each field been formatted consistently?
- Experiments:
 - Have units been used logically? If milliseconds have been used for some measurements and microseconds for others, is there a logical reason for doing so?
 - Has “megabyte” been written as “Mb” in some places and “Mbyte” in others?
 - Are all values of the same type presented with the same precision?
 - Are the graphs all the same size?
 - Are the axis units always given? If, say, the x-axes on different graphs measure the same units, do the axes have the same label?
 - Are all tables in the same format?
 - Are units given for every value?
 - Are labels and headings named consistently?
 - If, say, columns have been used for properties A to E in one table, have rows been used elsewhere? That is, do all tables have the same orientation?

- Each reference should be:
 - relevant;
 - up-to-date: check when taking over references from other papers;
 - reasonably accessible: pay attention to abbreviations of conference names, check validity of pointers to online material;
 - necessary.
- Three main styles:
 - numbered references: [16,32,18]
 - named references: (Chen and Li 2005, Deutsch et al. 1997)
 - uppercase abbreviations: [CL05, DPV97]

Which one to use is usually determined by the conference or format (IEEE, Springer, ACM, ...).
- Avoid making references the subject of a sentence:
 - No: [18] shows that query answering ...
 - Yes: Chen and Li [18] show that query answering ...
 - No: (Chen and Li 2005) shows that query answering ...
 - Yes: Chen and Li (2005) show that query answering ...